

TechMood DA — Lesson 04 PowerPivot and DAX Report

Executive conclusion

The business has generated **\$2,297,185.10 in sales** and **\$286,393.48 in profit**, with a **12.47% profit margin**. The most important margin-risk area is **Central**, whose regional margin is approximately **7.9%**. **Technology** is the strongest profit category.

Measures

Measure	DAX definition	Result
Total Sales	<code>Total Sales := SUM(Fact_Orders[Sales])</code>	\$2,297,185.10
Total Cost	<code>Total Cost := SUM(Fact_Orders[Total Cost])</code>	\$2,010,791.62
Total Profit	<code>Total Profit := SUM(Fact_Orders[Profit])</code>	\$286,393.48
Profit Margin %	<code>Profit Margin % := DIVIDE([Total Profit], [Total Sales], 0)</code>	12.47%
Average Order Value	<code>Average Order Value := DIVIDE([Total Sales], DISTINCTCOUNT(Fact_Orders[Order ID]), 0)</code>	\$458.70
Customer Count	<code>Customer Count := DISTINCTCOUNT(Fact_Orders[Customer ID])</code>	793

Data model and cost assumption

The workbook uses Fact_Orders with Dim_Return and Dim_People relationships. The supplied source did not include a physical Cost column, so Total Cost is derived as `Sales - Profit`. This is documented in the workbook so the assumption is transparent and auditable.

Pivot analysis by Category

Category	Total Sales	Total Cost	Total Profit	Profit Margin	AOV
Furniture	\$741,999.80	\$723,548.52	\$18,451.27	2.49%	\$420.63
Office Supplies	\$719,031.27	\$596,544.02	\$122,487.25	17.04%	\$192.15
Technology	\$836,154.03	\$690,699.08	\$145,454.95	17.40%	\$541.55

Pivot analysis by Region

Region	Total Sales	Total Cost	Total Profit	Profit Margin	AOV
Central	\$501,239.89	\$461,533.53	\$39,706.36	7.92%	\$426.59
East	\$678,765.48	\$587,246.25	\$91,519.23	13.48%	\$484.49
South	\$391,721.91	\$344,972.47	\$46,749.43	11.93%	\$476.55
West	\$725,457.82	\$617,039.38	\$108,418.45	14.94%	\$450.32

Three insights

1. **Technology is the most profitable category**, generating approximately \$145,454.95 profit and the strongest category margin.
2. **Central has the weakest regional margin**, approximately 7.92%, while West generates the highest regional profit. This indicates a margin-quality issue that should be investigated by discount, shipping, and product mix.
3. **Furniture is a margin-risk category**, generating only \$18,451.27 profit. Its costs and promotional policy should be reviewed before increasing sales volume.

Two recommendations

1. Introduce a minimum-margin approval rule for promotions and review Central-region orders plus Furniture products for discount, shipping, and return leakage.

2. Shift growth investment toward Technology and other high-margin products while monitoring Total Profit and Profit Margin % in every monthly Pivot review.

Screenshot

